EXPERIMENT 01 — SSH BRUTE-FORCE ATTACK DETECTION

BLUE TEAM **SOC LEVEL 1** **PRACTICAL INVESTIGATION** **2026/11/7**

1. What you will achieve

- By the end, you should be able to demonstrate:
    1. Simulating an SSH brute-force attack from Kali Linux.
    2. Recording failed authentication attempts on Ubuntu.
    3. Collecting authentication logs with Wazuh.
    4. Detecting repeated login failures.
    5. Investigating source IP, username, timestamps and rule severity.
    6. Producing a professional incident report.

2. Understand your existing lab

- Based on setup, our expected architecture is:

Machine	Role
Ubuntu	Wazuh Manager + Indexer + Dashboard + SSH target
Kali Linux	Attack simulation machine
Windows 11	Monitored endpoint

3. How detection works

i. KALI Linux – Generates failed SSH Login atttempts
ii. UBUNTU SSH Server – Records authentication failures in /var/log/auth.log
iii. Wazuh Log Collection – Collect and forwards local authentication events
iv. Wazuh Detection Engine – Decodes events and evaluates detection rules
v. SOC Analyst – investiges , classifies and report



Part A — Prepare the lab

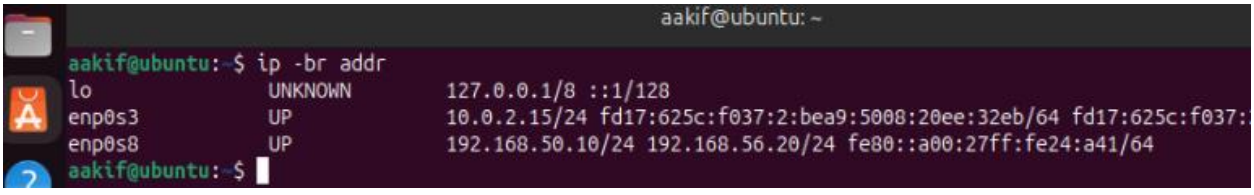
4. Step 1: Identify your Ubuntu configuration

Open your **Ubuntu** terminal.

Run these commands one by one.

Command 1 – Check IP addresses -192.168.50.10

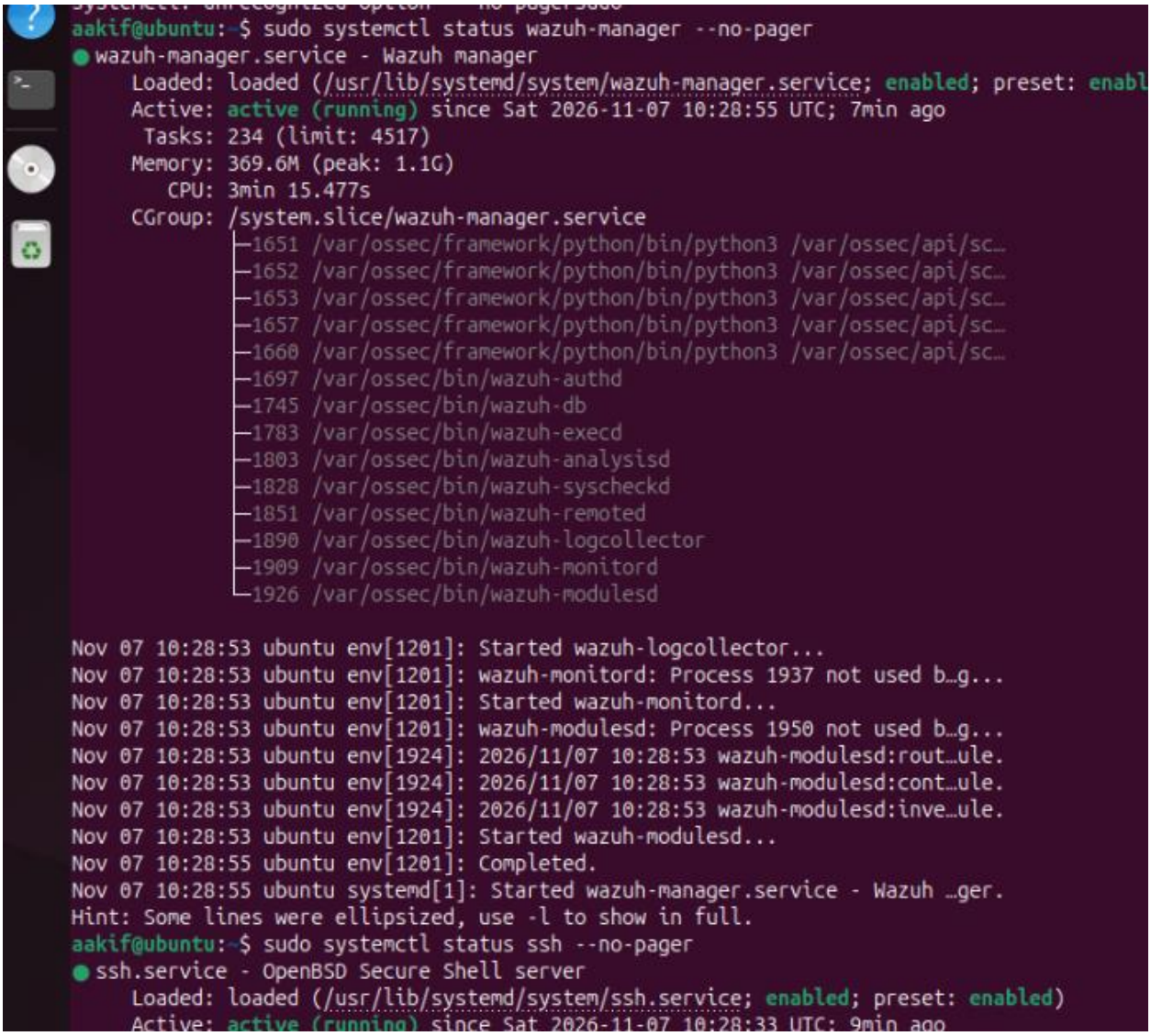
ip -br addr



Command 2 – Check Wazuh Manager

sudo systemctl status wazuh-manager --no-pager

result :



Command 3 – Check SSH service

sudo systemctl status ssh --no-pager

Result :

```
aakif@ubuntu:~$ sudo systemctl status ssh --no-pager
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-11-07 10:28:33 UTC; 9min ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 1168 (sshd)
      Tasks: 1 (limit: 4517)
     Memory: 1.3M (peak: 2.5M)
        CPU: 92ms
     CGroup: /system.slice/ssh.service
             └─1168 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Command 4 – Check authentication logs

sudo ls -lh /var/log/auth.log

Result :

```
aakif@ubuntu:~$
aakif@ubuntu:~$ sudo ls -lh /var/log/auth.log
-rw-r----- 1 syslog adm 239K Nov  7 10:39 /var/log/auth.log
```

Expected: a file listing with the log file size and permissions.

Then:

sudo tail -n 10 /var/log/auth.log

Result :

```
aakif@ubuntu:~$ sudo tail -n 10 /var/log/auth.log
2026-11-07T10:37:13.489078+00:00 ubuntu pkexec: pam_unix(polkit-1:session): session opened for user root(uid=0) by aakif(uid=1000)
2026-11-07T10:37:13.499546+00:00 ubuntu pkexec[18025]: aakif: Executing command [USER=root] [TTY=unknown] [CWD=/home/aakif] [COMMAND=/usr/lib/update-notifier/package-system-locked]
2026-11-07T10:38:31.071088+00:00 ubuntu sudo:    aakif : TTY=pts/0 ; PWD=/home/aakif ; USER=root ; COMMAND=/usr/bin/systemctl status ssh --no-pager
2026-11-07T10:38:31.084296+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by aakif(uid=1000)
2026-11-07T10:38:40.233191+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-11-07T10:39:28.935044+00:00 ubuntu sudo:    aakif : TTY=pts/0 ; PWD=/home/aakif ; USER=root ; COMMAND=/usr/bin/ls -lh /var/log/auth.log
2026-11-07T10:39:28.953302+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by aakif(uid=1000)
2026-11-07T10:39:28.960064+00:00 ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-11-07T10:39:53.203035+00:00 ubuntu sudo:    aakif : TTY=pts/0 ; PWD=/home/aakif ; USER=root ; COMMAND=/usr/bin/tail -n 10 /var/log/auth.log
2026-11-07T10:39:53.220676+00:00 ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by aakif(uid=1000)
aakif@ubuntu:~$
```

You should see recent authentication messages if SSH or other authentication activity has been recorded.

5. Step 2: Check Wazuh's local log collection

On Ubuntu:

sudo grep -n -A 4 -B 2 "auth.log" /var/ossec/etc/ossec.conf

Result :



- We want to find a configuration similar to:

<localfile>

<log_format>syslog</log_format>

<location>/var/log/auth.log</location>

</localfile>

If it exists, Wazuh is configured to collect that authentication log.

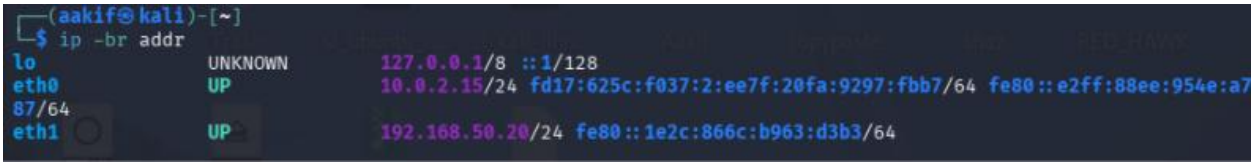
If it does not exist, we will add the configuration carefully in the next stage rather than blindly editing the file.

6. Step 3: Identify Kali's IP address

Open Kali Linux and run:

ip -br addr

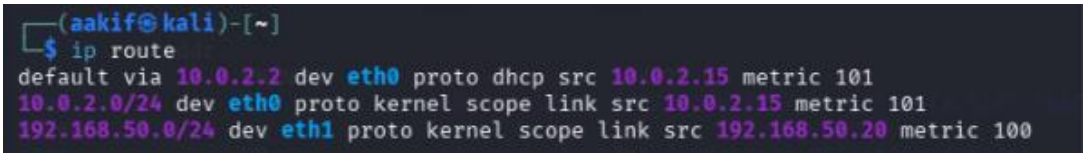
Result : 192.168.50.20/24



Now identify the route to Ubuntu:

ip route

Result :



7. Step 4: Verify SSH connectivity

From Kali:

nc -vz 192.168.50.10 22

Result :

```
┌──(aakif㉿kali)-[~/wazuh-ssh-lab]
└─$ nc -vz 192.168.50.10 22
Connection to 192.168.50.10 22 port [tcp/ssh] succeeded!
```

Part B — Simulate the attack

8. Step 5: Prepare Hydra

On Kali:

sudo apt update

sudo apt install hydra -y

Result :

```
┌──(aakif㉿kali)-[~]
└─$ sudo apt update
sudo apt install hydra -y
Get:1 http://mirror.kku.ac.th/kali kali-rolling InRelease [34.0 kB]
Get:2 http://mirror.kku.ac.th/kali kali-rolling/main amd64 Packages [21.9 MB]
Get:3 http://mirror.kku.ac.th/kali kali-rolling/main amd64 Contents (deb) [54.5 MB]
Get:4 http://mirror.kku.ac.th/kali kali-rolling/non-free amd64 Packages [170 kB]
Fetched 76.6 MB in 10min 37s (120 kB/s)
2143 packages can be upgraded. Run 'apt list --upgradable' to see them.
hydra is already the newest version (9.7-2).
hydra set to manually installed.
The following packages were automatically installed and are no longer required:
  bloodhound.py  libavformat61           libmjpegutils-2.1-0t64  libpocketsphinx3   libvdpau-va-gl1     ruby-unf-ext
  curlftpfs      libconfig-inifiles-perl  libmpeg2encpp-2.1-0t64  libpostproc58      mesa-vdpau-drivers  vdpau-driver-all
  libaudio2      libfuse2t64             libmplex2-2.1-0t64      libsphinxbase3t64  pocketsphinx-en-us
  libavfilter10  libgav1-1               libmupdf25.1            libswscale8        python3-fs
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2143
```

Create a dedicated lab directory:

mkdir -p ~/wazuh-ssh-lab

cd ~/wazuh-ssh-lab

Result :

```
┌──(aakif㉿kali)-[~]
└─$ mkdir -p ~/wazuh-ssh-lab
cd ~/wazuh-ssh-lab
```

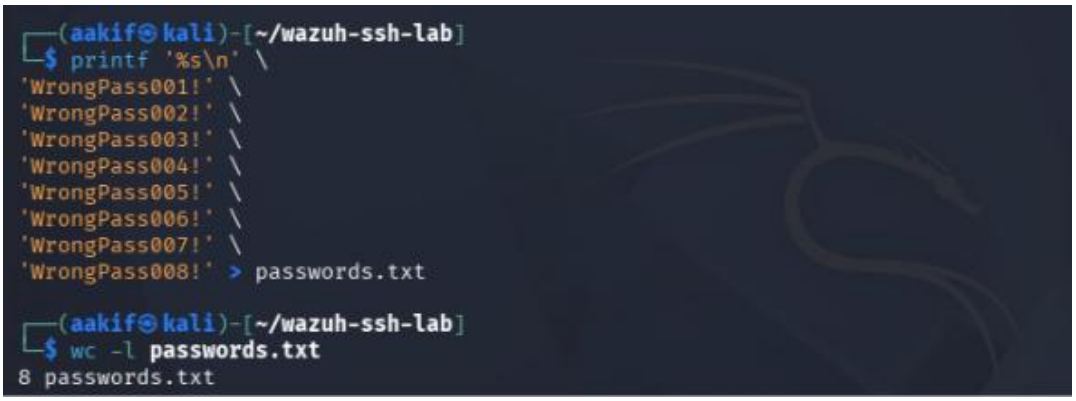
Create eight deliberately incorrect passwords:

*printf '%s\n' *
*'WrongPass001!' *
*'WrongPass002!' *
*'WrongPass003!' *
*'WrongPass004!' *
*'WrongPass005!' *
*'WrongPass006!' *
*'WrongPass007!' *
'WrongPass008!' > passwords.txt

Verify:

wc -l passwords.txt

Result :



9. Step 6: Monitor the target while attacking

Open two terminals.

Terminal 1 – Ubuntu

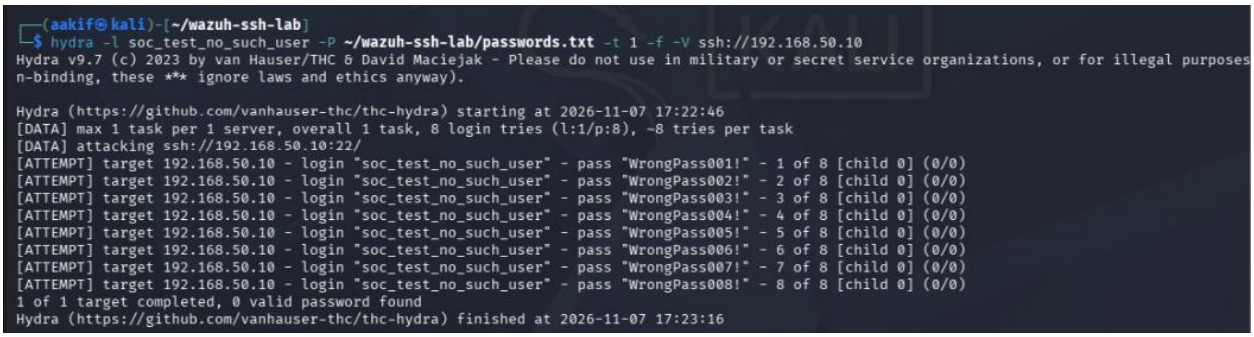
sudo tail -f /var/log/auth.log

Terminal 2 – Kali

Once all previous checks are successful, run:

hydra -l soc_test_no_such_user -P ~/wazuh-ssh-lab/passwords.txt -t 1 -f -V ssh://192.168.50.10

Expected Hydra output :



Part C — Investigate Wazuh

10. Step 7: Verify the alert on Ubuntu

On your Wazuh Manager machine:

sudo grep '"id":"5712"' /var/ossec/logs/alerts/alerts.json | tail -n 5